Find the Error

5.OA.1

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate (3 + 4) × 2. Student's answer: 11

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?

The student multiplied 4 × 2 before working inside the parentheses. Do (3 + 4) first: 7 × 2 = 14.

Solve the problem correctly. Show all of your work.

14

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 20 - (5 + 3). Student's answer: 18

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did 20 - 5 + 3. Work inside the parentheses first: 5 + 3 = 8, then 20 - 8 = 12.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 6 × (2 + 4). Student's answer: 16

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did 6 × 2 + 4. Parentheses come first: 2 + 4 = 6, then 6 × 6 = 36.	**36**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 24 ÷ (2 + 6). Student's answer: 18

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 24 by 2 and then added 6. Add inside the parentheses first: 2 + 6 = 8, then 24 ÷ 8 = 3.	**3**

5

Find the Error

5.OA.1

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate [(10 - 2) × 3]. Student's answer: 4

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?

The student subtracted 2 from 10 and then did the multiplication wrong. Start with 10 - 2 = 8, then 8 × 3 = 24.

Solve the problem correctly. Show all of your work.

24

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 5 + 2 × (7 - 3). Student's answer: 28

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 5 + 2 before multiplying. Multiply before adding: 2 × 4 = 8, then 5 + 8 = 13.	**13**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: add 6 to 4, then multiply by 3. Student writes: 6 + 4 × 3

Write the expression with parentheses.

What did the student do wrong? Why do you think the student made this error?

Parentheses are needed so the sum is multiplied by 3. The correct expression is (4 + 6) × 3.

Solve the problem correctly. Show all of your work.

(4 + 6) × 3

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: subtract 5 from 20, then divide by 3. Student writes: 20 - 5 ÷ 3

Write the expression with parentheses.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Without parentheses, only 5 is divided by 3. The correct expression is (20 - 5) ÷ 3.	**(20 - 5) ÷ 3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: twice the sum of 7 and 1. Student writes: 2 + 7 + 1

Write the expression.

What did the student do wrong? Why do you think the student made this error?

Twice means multiply by 2, and the sum needs parentheses. The correct expression is 2 × (7 + 1).

Solve the problem correctly. Show all of your work.

2 × (7 + 1)

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write an expression for: three less than 10, then multiplied by 4. Student writes: $10 - 3 \times 4$

Write the expression with parentheses.

What did the student do wrong? Why do you think the student made this error?

The student multiplied 3 by 4 first. Use parentheses so the difference is multiplied: (10 - 3) × 4.

Solve the problem correctly. Show all of your work.

(10 – 3) × 4

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without calculating, is $3 \times (4 + 5)$ equal to $3 \times 4 + 5$? Student says: Yes

Decide yes or no and explain.

What did the student do wrong? Why do you think the student made this error?

They are not equal. $3 \times 9 = 27$, but $3 \times 4 + 5 = 17$. Parentheses change the order of operations.

Solve the problem correctly. Show all of your work.

No, 27 does not equal 17

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 2 × (8 - 3). Student's answer: 16

Evaluate the expression.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 2 by 8 and skipped the 3. Work inside the parentheses first: 8 - 3 = 5, then 2 × 5 = 10.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The pattern starts at 3 and adds 4 each time: 3, 7, 11, 15, ... Find the 5th term. Student's answer: 15

Continue the pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student repeated the 4th term. Add 4 again: 15 + 4 = 19.	**19**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

attern A is 0, 3, 6, 9, … Pattern B is 0, 6, 12, 18, … Compare the matching terms. Student says: B is 3 more than

Compare the matching terms.

What did the student do wrong? Why do you think the student made this error?

The student compared the differences between terms. Compare matching terms: 6 ÷ 3 = 2 and 12 ÷ 6 = 2, so each term in B is twice the matching term in A.

Solve the problem correctly. Show all of your work.

Each term in B is twice as large as the matching term in A

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The pattern starts at 20 and subtracts 2 each time. Find the 5th term. Student's answer: 10

Continue the pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 4 from 14 instead of 2. The pattern is 20, 18, 16, 14, 12, so the 5th term is 12.	**12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = 4 × x. When x = 3, what is y? Student's answer: 7

Use the rule y = 4 × x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 instead of multiplying. Use the rule: 4 × 3 = 12.	**12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has y = x + 5. When x = 4, what is y? Student's answer: 20

Use the rule y = x + 5.

What did the student do wrong? Why do you think the student made this error?

The student multiplied 4 × 5. The rule adds 5: 4 + 5 = 9.

Solve the problem correctly. Show all of your work.

9

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ben says the 10th term of the pattern 4, 8, 12, … is 36. Is he right? Student says: Yes

Check the 10th term.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Ben used the 9th term, 9 × 4 = 36. The 10th term is 10 × 4 = 40, so he is not right.	**No, the 10th term is 40**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In 2,500, what is the value of the 5? Student's answer: 5

Find the place value of the digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 5 is in the hundreds place, so it is worth 5 hundreds, or 500.	**500**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In 8,042, what is the value of the 8? Student's answer: 80

Find the place value of the digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 8 is in the thousands place, so it is worth 8,000.	**8,000**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 6 in the hundreds place is how many times as large as the 6 in the tens place? Student's answer: 100 time

Compare the places.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each place to the left is 10 times the place to its right. Hundreds is 10 times tens.	**10 times**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is 10 times as large as 600: 60 or 6,000? Student's answer: 60

Multiply by 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student chose the smaller number. 10 × 600 = 6,000.	**6,000**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is one tenth of 3,000? Student's answer: 30

Divide by 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided by 100. One tenth means divide by 10: 3,000 ÷ 10 = 300.	**300**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 4,000 + 20 + 7 in standard form. Student writes: 4,207

Write the number in standard form.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student put the 2 in the hundreds place. The 20 belongs in the tens place, so the number is 4,027.	**4,027**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 10^2? Student's answer: 20

Evaluate the power of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The exponent tells how many 10s to multiply: 10 × 10 = 100.	**100**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 3×10^3? Student's answer: 300

Evaluate the power of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
10^3 means $10 \times 10 \times 10 = 1{,}000$, not 100. So $3 \times 1{,}000 = 3{,}000$.	**3,000**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 7.25 × 100? Student's answer: 72.5

Move the decimal point.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Multiplying by 100 moves the decimal point two places to the right, not one. The answer is 725.	**725**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 10 as a standard number. Student writes: 40

Write the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 10 by 4. 10 means 10 × 10 × 10 × 10 = 10,000.	**10,000**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 1,200 ÷ 100? Student's answer: 120

Divide by a power of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Dividing by 100 moves the decimal point two places to the left. 1,200 ÷ 100 = 12.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What exponent makes 10 to that power equal 1,000,000? Student's answer: 5

Count the zeros.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
1,000,000 has six zeros, so it equals 10.	**6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.304 in words. Student writes: three hundred four

Write the decimal in words.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The last digit is in the thousandths place, so the words end with thousandths: three hundred four thousandths.	**three hundred four thousandths**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is greater: 0.45 or 0.405? Student's answer: 0.405

Compare the decimals.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Line up the decimals: 0.450 is greater than 0.405 because 0.45 has more tenths and hundredths.	**0.45**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 2.614 in expanded form. Student writes: 2 + 6/10 + 4/100

Write the expanded form.

What did the student do wrong? Why do you think the student made this error?

The digit 4 is in the thousandths place, so it is 4/1,000. The student wrote 4/100, which is a different value. Correct: 2 + 6/10 + 1/100 + 4/1,000.

Solve the problem correctly. Show all of your work.

2 + 6/10 + 1/100 + 4/1,000

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is greater: 3.8 or 3.080? Student's answer: 3.080

Compare the decimals.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Write 3.8 as 3.800. Compare 3.800 and 3.080. The tenths digit tells you 3.8 is greater.	**3.8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write seven and nine thousandths as a decimal. Student writes: 7.09

Write the decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Nine thousandths has 9 in the thousandths place, so the decimal is 7.009.	**7.009**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Order from least to greatest: 0.6, 0.06, 0.606. Student writes: 0.06, 0.606, 0.6

Order the decimals.

What did the student do wrong? Why do you think the student made this error?

Write the numbers with the same places: 0.060, 0.600, 0.606. The correct order is 0.06, 0.6, 0.606.

Solve the problem correctly. Show all of your work.

0.06, 0.6, 0.606

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 3.456 to the nearest hundredth. Student's answer: 3.45

Round the number.

What did the student do wrong? Why do you think the student made this error?

Look at the thousandths digit, 6. Since 6 is 5 or more, round the hundredths digit up: 3.46.

Solve the problem correctly. Show all of your work.

3.46

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 6.282 to the nearest tenth. Student's answer: 6.2

Round the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Look at the hundredths digit, 8. Since 8 is 5 or more, round the tenths digit up: 6.3.	**6.3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 14.5 to the nearest whole number. Student's answer: 14

Round the number.

What did the student do wrong? Why do you think the student made this error?

The tenths digit is 5, so round up. The nearest whole number is 15.

Solve the problem correctly. Show all of your work.

15

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 9.05 to the nearest tenth. Student's answer: 9.0

Round the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The hundredths digit is 5, so round the tenths digit up. 9.05 rounds to 9.1.	**9.1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 0.7384 to the nearest hundredth. Student's answer: 0.73

Round the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The thousandths digit is 8, so round up. 0.7384 rounds to 0.74.	**0.74**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 12.996 to the nearest tenth. Student's answer: 12.9

Round the number.

What did the student do wrong? Why do you think the student made this error?

The hundredths digit is 9, so round the tenths digit up. 9 + 1 = 10, which carries. The answer is 13.0.

Solve the problem correctly. Show all of your work.

13.0

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 36 × 24. Student's answer: 764

Use partial products.

What did the student do wrong? Why do you think the student made this error?

The partial products are 144 and 720. Add them: 144 + 720 = 864.

Solve the problem correctly. Show all of your work.

864

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 207 × 5. Student's answer: 135

Multiply each part.

What did the student do wrong? Why do you think the student made this error?

The student multiplied only 7 × 5. Multiply 200 × 5 = 1,000 too. The total is 1,035.

Solve the problem correctly. Show all of your work.

1,035

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 48 × 50. Student's answer: 240

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 50 has 5 tens, so the answer has an extra zero. 48 × 5 = 240, then × 10 = 2,400.	**2,400**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 125 × 16. Student's answer: 1,925

Use partial products.

What did the student do wrong? Why do you think the student made this error?

The partial products are 125 × 6 = 750 and 125 × 10 = 1,250. Add them: 2,000.

Solve the problem correctly. Show all of your work.

2,000

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 73 × 9. Student's answer: 647

Multiply and add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
9 × 73 = 9 × 70 + 9 × 3 = 630 + 27 = 657. The student added the carried ten wrong.	**657**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 12 boxes with 35 crayons each. How many crayons in all? Student's answer: 415

Multiply and check.

What did the student do wrong? Why do you think the student made this error?

12 × 30 = 360 and 12 × 5 = 60. Add them: 360 + 60 = 420.

Solve the problem correctly. Show all of your work.

420

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 384 ÷ 12. Student's answer: 34

Divide and check with multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Check: 12 × 34 = 408, which is too big. 12 × 32 = 384, so the answer is 32.	**32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 936 ÷ 24. Student's answer: 40

Estimate, then divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
24 × 40 = 960, which is too big. 24 × 39 = 936, so the answer is 39.	**39**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 1,750 ÷ 25. Student's answer: 7

Divide.

What did the student do wrong? Why do you think the student made this error?

25 × 7 = 175. Since 1,750 is 10 times as large, the answer is 70.

Solve the problem correctly. Show all of your work.

70

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 257 ÷ 16. Student's answer: 16 remainder 9

Divide and find the remainder.

What did the student do wrong? Why do you think the student made this error?

16 × 16 = 256. Subtract: 257 − 256 = 1. The answer is 16 remainder 1.

Solve the problem correctly. Show all of your work.

16 remainder 1

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 3,528 ÷ 42. Student's answer: 74

Divide and check with multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Check: 42 × 74 = 3,108, which is too small. 42 × 80 = 3,360 and 42 × 4 = 168. Together that is 3,528, so the answer is 84.	**84**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

360 stickers are shared equally among 15 kids. How many does each kid get? Student's answer: 26

Divide and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
15 × 26 = 390, which is too many. 15 × 24 = 360, so each kid gets 24.	**24**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 2.4 + 3.75. Student's answer: 5.15

Line up the decimal points.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Line up the decimal points and add 2.40 + 3.75. The answer is 6.15.	**6.15**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 8.5 - 3.27. Student's answer: 5.37

Line up the decimal points and subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Write 8.50. You need to borrow from the tenths place for the hundredths. The answer is 5.23.	**5.23**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 0.6 × 0.4. Student's answer: 2.4

Multiply as whole numbers, then place the decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 × 4 = 24. The two factors have two decimal places in total, so the answer is 0.24.	**0.24**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 4.8 ÷ 6. Student's answer: 8

Divide and place the decimal point.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The decimal point was dropped. Put the decimal point in the quotient above the one in 4.8: 0.8.	**0.8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 3.6 × 100. Student's answer: 36

Move the decimal point.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Multiplying by 100 moves the decimal point two places to the right. The answer is 360.	**360**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 7.5 ÷ 3. Student's answer: 2.05

Divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
7 ÷ 3 = 2 remainder 1. Bring down the 5 to make 15, and 15 ÷ 3 = 5. The answer is 2.5.	**2.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 1/2 + 1/3. Student's answer: 2/5

Find a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the numerators and the denominators. Use the common denominator 6: 3/6 + 2/6 = 5/6.	**5/6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 3/4 - 1/6. Student's answer: 2/2

Find a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted the denominators too. Use 12: 9/12 - 2/12 = 7/12.	**7/12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 2/3 + 3/5. Student's answer: 5/8

Find a common denominator and simplify.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the tops and the bottoms. Use 15: 10/15 + 9/15 = 19/15, which is 1 4/15.	**1 4/15**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 5/6 - 1/4. Student's answer: 4/2

Find a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not change to a common denominator first. Use 12: 10/12 - 3/12 = 7/12.	**7/12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 7/8 + 1/4. Student's answer: 8/12

Find a common denominator and simplify.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Change 1/4 to 2/8: 7/8 + 2/8 = 9/8, which is 1 1/8.	**1 1/8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 3/5 + 1/3. Student's answer: 4/8

Find a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the tops and bottoms. Use 15: 9/15 + 5/15 = 14/15.	**14/15**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Maria ate 1/3 of a pizza. Leo ate 1/4 of the same pizza. How much did they eat in all? Student's answer: 2/7

Use a common denominator to add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Use 12: 4/12 + 3/12 = 7/12.	**7/12 of the pizza**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bottle has 3/4 liter of juice. Kai pours out 1/2 liter. How much is left? Student's answer: 1/2 liter

Subtract to find what is left.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
1/2 liter is what was poured out, not what is left. 3/4 - 1/2 = 3/4 - 2/4 = 1/4 liter.	**1/4 liter**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without finding the exact sum, is 5/9 + 7/10 more or less than 1? Student says: less than 1

Estimate with benchmark fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
5/9 is a little more than 1/2, and 7/10 is more than 1/2. Together they are more than 1, so the sum is more than 1.	**More than 1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ann walks 2/5 mile, then 1/3 mile. How far does she walk in all? Student's answer: 3/8 mile

Use a common denominator to add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Use 15: 6/15 + 5/15 = 11/15 mile.	**11/15 mile**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A recipe uses 2 1/4 cups of flour and 1 1/2 cups of sugar. What is the total? Student's answer: 3 2/4

Add the whole numbers and the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted the fractions. 1/4 + 1/2 = 1/4 + 2/4 = 3/4, so the total is 3 3/4 cups.	**3 3/4 cups**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Sam hikes 4 1/2 miles of a 6-mile trail. How much of the trail is left? Student's answer: 2 1/2 miles

Subtract to find what is left.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 - 4 1/2 = 1 1/2 miles. The student left out the half when subtracting.	**1 1/2 miles**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 3 ÷ 4 as a fraction. Student writes: 4/3

Write the division as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The number being divided goes on top. 3 ÷ 4 = 3/4.	**3/4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 sandwiches are shared equally by 2 people. How much does each person get? Student's answer: 2/5

Write the division and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the numbers. 5 ÷ 2 = 5/2 = 2 1/2 sandwiches each.	**2 1/2 sandwiches**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 7/3 as a mixed number. Student's answer: 3 1/3

Divide the numerator by the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
7 ÷ 3 = 2 remainder 1. So 7/3 = 2 1/3.	**2 1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 4 ÷ 9 equal to 9/4? Student says: Yes

Decide yes or no.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The fraction 4/9 is 4 ÷ 9. The student wrote the reverse, 9/4.	**No, 4 ÷ 9 = 4/9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

liter bottle of juice is shared equally among 5 cups. How much juice is in each cup? Student's answer: 5/2 l

Write the division and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the numbers. 2 ÷ 5 = 2/5 liter in each cup.	**2/5 liter**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ounds of rice are shared equally among 4 bags. How much rice is in each bag? Student's answer: 4/3 pour

Write the division and solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 4 by 3. Divide 3 by 4: each bag gets 3/4 pound.	**3/4 pound**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 2/3 × 3/5. Student's answer: 6/8

Multiply the numerators and the denominators.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Multiply: 2 × 3 = 6 and 3 × 5 = 15, so 6/15. Simplify to 2/5.	**2/5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is 3/4 foot wide and 2/3 foot long. What is its area? Student's answer: 5/7 square foot

Multiply the length and width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Area is length times width, so multiply: 3/4 × 2/3 = 6/12 = 1/2 square foot.	**1/2 square foot**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is 3/4 of 12? Student's answer: 36

Multiply the fraction by the whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 3 × 12 and kept the 4 as the denominator. Divide 12 by 4 first, then multiply by 3: 9.	**9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 2 1/2 × 3 1/3. Student's answer: 6 1/6

Change to improper fractions and multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Change to improper fractions: 5/2 × 10/3 = 50/6 = 8 1/3. The student multiplied the whole numbers and fractions separately.	**8 1/3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Jen has 3/4 yard of ribbon and uses 2/3 of it. How much ribbon did she use? Student's answer: 5/7 yard

Multiply the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the fractions. Multiply: 2/3 × 3/4 = 6/12 = 1/2 yard.	**1/2 yard**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle is 1 1/2 feet by 2 feet. What is its area? Student's answer: 2 1/2 square feet

Multiply the length and width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides. Area is length times width: 3/2 × 2 = 3 square feet.	**3 square feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without multiplying, is 6 × 2/3 larger or smaller than 6? Student says: larger

Decide if the product is larger or smaller.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2/3 is less than 1, so multiplying by it makes the number smaller. The answer is 4, which is smaller than 6.	**Smaller**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without multiplying, is 5 × 3/2 larger or smaller than 5? Student says: smaller

Decide if the product is larger or smaller.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3/2 is greater than 1, so multiplying by it makes the number larger. The answer is 7 1/2.	**Larger**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without multiplying, is 9 × 5/5 equal to 9, larger than 9, or smaller than 9? Student says: larger

Compare the factor to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
5/5 equals 1, so the product stays the same. 9 × 5/5 = 9.	**Equal to 9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is larger: 4 × 5/3 or 4 × 2/3? Student says: 4 × 2/3

Compare the factors.

What did the student do wrong? Why do you think the student made this error?

5/3 is greater than 2/3, so 4 × 5/3 is larger. Multiplying by a bigger factor gives a bigger product.

Solve the problem correctly. Show all of your work.

4 × 5/3

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without multiplying, is 9 × 2/5 larger or smaller than 9? Student says: larger

Decide if the product is larger or smaller.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2/5 is less than 1, so the product is smaller than 9.	**Smaller**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Without multiplying, is 25 × 1 1/4 larger or smaller than 25? Student says: smaller

Decide if the product is larger or smaller.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
1 1/4 is greater than 1, so the product is larger than 25.	**Larger**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s 12 feet long. Ben cuts off 1/4 of it. Then he cuts off 1/3 of the rest. How long is the piece left? Student's answe

Solve each step.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student stopped after the first cut. After the first cut, 9 feet is left. Then 1/3 of 9 is 3 feet, so 6 feet is left.	**6 feet**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A hiker walks 2/5 mile, then 3/10 mile, then 1/2 mile. How far does she walk in all? Student's answer: 2 1/5 mile

Use a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The fractions in tenths are 4/10, 3/10, and 5/10. They add to 12/10, which is 1 2/10 or 1 1/5 miles.	**1 1/5 miles**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

apples. 1/4 of them are red. Half of the remaining apples are green. How many green apples are there? Stu

Solve each step.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student took half of all 24 apples. First find the remaining apples: 24 - 6 = 18. Half of 18 is 9.	**9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

gallons and is 3/8 full. Then 10 more gallons are added. How many gallons are in the tank now? Student's a

Solve each step.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to add the 10 gallons. 3/8 of 40 is 15, and 15 + 10 = 25 gallons.	**25 gallons**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shirt costs $36. It is on sale for 1/3 off. What is the sale price? Student's answer: $12

Find the discount, then the sale price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$12 is the amount saved, not the sale price. $36 - $12 = $24.	**$24**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

/5 of a book on Monday. On Tuesday she reads half of the rest. What fraction of the book is left? Student's c

Solve each step.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student took half of the whole book instead of half of the 4/5 left. Half of 4/5 is 2/5, so 4/5 - 2/5 = 2/5 is left.	**2/5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 1/3 ÷ 2. Student's answer: 2/3

Divide the unit fraction by the whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 2. Dividing 1/3 into 2 equal parts gives 1/6.	**1/6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 4 ÷ 1/5. Student's answer: 4/5

Ask how many 1/5s fit in 4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 5 fifths in each whole, so there are 4 × 5 = 20 fifths in 4 wholes.	**20**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 1/2 ÷ 4. Student's answer: 2

Divide the unit fraction by the whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the other way. Split 1/2 into 4 equal parts: 1/8.	**1/8**

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Find the Error

5.NF.7

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide 3 ÷ 1/4. Student's answer: 3/4

Ask how many 1/4s fit in 3.

What did the student do wrong? Why do you think the student made this error?

Each whole has 4 fourths, so 3 wholes have 3 × 4 = 12 fourths.

Solve the problem correctly. Show all of your work.

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Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many 1/6-cup servings are in 2 cups of soup? Student's answer: 3

Ask how many 1/6s fit in 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each cup has 6 sixths. So 2 cups have 2 × 6 = 12 servings.	**12 servings**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s 1/2 gallon of paint. She pours it equally into 5 cups. How much paint is in each cup? Student's answer: 5/2

Divide the fraction by the whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the numbers. 1/2 ÷ 5 = 1/10 gallon in each cup.	**1/10 gallon**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 3 feet to inches. Student's answer: 30 inches

Use the conversion for feet and inches.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 12 inches in 1 foot. So 3 × 12 = 36 inches.	**36 inches**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 4 kilograms to grams. Student's answer: 400 grams

Use the conversion for kilograms and grams.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 1,000 grams in 1 kilogram. So 4 × 1,000 = 4,000 grams.	**4,000 grams**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 2.5 liters to milliliters. Student's answer: 250 milliliters

Use the conversion for liters and milliliters.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 1,000 milliliters in 1 liter. So 2.5 × 1,000 = 2,500 milliliters.	**2,500 milliliters**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 3 hours 45 minutes to minutes. Student's answer: 345 minutes

Use 60 minutes in each hour.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student wrote the digits together. 3 hours = 180 minutes, and 180 + 45 = 225 minutes.	**225 minutes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Convert 3 quarts to cups. Student's answer: 6 cups

Use 4 cups in each quart.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 4 cups in 1 quart. So 3 × 4 = 12 cups.	**12 cups**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

2 meter rope is cut into 5 equal pieces. How long is each piece in centimeters? Student's answer: 250 centi

Convert to centimeters, then divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2.5 meters is 250 centimeters, but the student did not split it into 5 pieces. 250 ÷ 5 = 50 centimeters.	**50 centimeters**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ot shows these lengths in inches: 1/4, 1/2, 1/2, 3/4, 1, 1/4. What is the total of all the lengths? Student's answer:

Add all the measurements.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The total is 1/4 + 1/2 + 1/2 + 3/4 + 1 + 1/4 = 3 1/4 inches. The student left out the fraction.	**3 1/4 inches**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ne plot shows 1/2, 1/2, 3/4, and 1 1/4 inches. How many measurements are less than 1 inch? Student's answe

Count the measurements less than 1 inch.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The measurements 1/2, 1/2, and 3/4 are all less than 1 inch, so the count is 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

bean sprouts grew these amounts in inches: 1/8, 1/4, 1/4, 3/8. What is the total growth? Student's answer: 1/2

Add with a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Use eighths: 1/8 + 2/8 + 2/8 + 3/8 = 8/8 = 1 inch.	**1 inch**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

t shows 1/2, 1 1/4, and 3/4 inches. What is the difference between the largest and smallest? Student's answe

Subtract the smallest from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The largest is 1 1/4 and the smallest is 1/2. 1 1/4 - 1/2 = 3/4 inch.	**3/4 inch**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ribbon pieces measure 3/8, 3/8, 5/8, and 7/8 inch. What is the total length? Student's answer: 1 1/4 inches

Add the lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Add the eighths: 3 + 3 + 5 + 7 = 18, so the total is 18/8 = 2 2/8 = 2 1/4 inches.	**2 1/4 inches**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The data is 3/4, 1/2, 3/4, 1, and 3/4 inch. How many measurements equal 3/4 inch? Student's answer: 2

Count the matching measurements.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Three of the measurements are 3/4 inch, so the count is 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cube has sides that are each 1 unit long. What is its volume? Student's answer: 1 square unit

Choose the correct unit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Volume is measured in cubic units because it counts cubes. The answer is 1 cubic unit.	**1 cubic unit**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box has 3 layers of 8 unit cubes each. What is its volume? Student's answer: 11 cubic units

Find the total number of cubes.

What did the student do wrong? Why do you think the student made this error?

The student added 3 + 8. Multiply the layers by the cubes in each layer: 3 × 8 = 24 cubic units.

Solve the problem correctly. Show all of your work.

24 cubic units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which unit measures volume: square centimeter or cubic centimeter? Student's answer: square centimeter

Choose the correct unit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Square units measure area. Volume uses cubic units, such as cubic centimeters.	**Cubic centimeter**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cube has sides that are each 2 units long. What is its volume? Student's answer: 4 cubic units

Multiply the three side lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the area of one face. Volume is 2 × 2 × 2 = 8 cubic units.	**8 cubic units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box has 5 layers of 6 unit cubes each. How many cubes fill the box? Student's answer: 6

Count all the cubes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only one layer. There are 5 × 6 = 30 cubes.	**30 cubes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 2 units by 2 units by 2 units. How many unit cubes fill it? Student's answer: 12

Count the cubes in each layer, then the layers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the edges. There are 4 cubes in each layer and 2 layers, so 2 × 2 × 2 = 8 cubes.	**8 cubes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

solid has 4 layers, and each layer is 3 by 2 unit cubes. How many cubes are in the solid? Student's answer:

Count the cubes in each layer, then the layers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each layer has 3 × 2 = 6 cubes. With 4 layers, the total is 4 × 6 = 24 cubes.	**24 cubes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

has 2 layers, and each layer is a 3 by 3 square of unit cubes. How many cubes are in the solid? Student's ar

Count the cubes in each layer, then the layers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only one layer. Each layer has 9 cubes, so 2 layers have 18 cubes.	**18 cubes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ism is built with unit cubes. It is 5 cubes long, 2 cubes wide, and 3 cubes tall. How many cubes are in it? Stu

Multiply the three dimensions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied only two dimensions. 5 × 2 × 3 = 30 cubes.	**30 cubes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

base is 4 cubes by 5 cubes. The stack is 3 cubes high. How many cubes are in the stack? Student's answer:

Multiply the base by the height.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only the base. The base has 20 cubes, and 3 layers give 3 × 20 = 60 cubes.	**60 cubes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 6 cubes long, 2 cubes wide, and 4 cubes tall. How many cubes fill it? Student's answer: 12

Multiply the three dimensions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 6 × 2 only. Multiply by the height too: 6 × 2 × 4 = 48 cubes.	**48 cubes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e layer has 8 cubes arranged 2 by 4. There are 5 layers. How many cubes are there in all? Student's answer

Multiply the cubes in a layer by the number of layers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 8 + 5. Multiply instead: 8 × 5 = 40 cubes.	**40 cubes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 4 cm long, 3 cm wide, and 2 cm tall. What is its volume? Student's answer: 9 cubic cm

Use V = l × w × h.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides. Multiply: 4 × 3 × 2 = 24 cubic cm.	**24 cubic cm**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 6 units long, 5 units wide, and 2 units tall. What is its volume? Student's answer: 22 cubic units

Use V = l × w × h.

What did the student do wrong? Why do you think the student made this error?

The student added 6 + 5 and then multiplied by 2. Multiply all three: 6 × 5 × 2 = 60 cubic units.

Solve the problem correctly. Show all of your work.

60 cubic units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 1.5 units long, 2 units wide, and 4 units tall. What is its volume? Student's answer: 7.5 cubic units

Use V = l × w × h.

What did the student do wrong? Why do you think the student made this error?

The student added the sides or made a multiplication mistake. 1.5 × 2 = 3, and 3 × 4 = 12 cubic units.

Solve the problem correctly. Show all of your work.

12 cubic units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

box has a volume of 48 cubic units. It is 4 units long and 3 units wide. How tall is it? Student's answer: 8 unit

Use the missing dimension.

What did the student do wrong? Why do you think the student made this error?

The student divided by 6, not 12. 4 × 3 = 12, and 48 ÷ 12 = 4 units tall.

Solve the problem correctly. Show all of your work.

4 units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 10 units long, 4 units wide, and 3 units tall. What is its volume? Student's answer: 17 cubic units

Use V = l × w × h.

What did the student do wrong? Why do you think the student made this error?

The student added the three sides.
Multiply: 10 × 4 × 3 = 120 cubic units.

Solve the problem correctly. Show all of your work.

120 cubic units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tank is 20 inches long, 10 inches wide, and 12 inches tall. What is its volume? Student's answer: 240 cubic in

Use V = l × w × h.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student dropped a zero. 20 × 10 = 200, and 200 × 12 = 2,400 cubic inches.	**2,400 cubic inches**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the x-coordinate of the point (3, 5)? Student's answer: 5

Identify the x-coordinate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The first number in an ordered pair is the x-coordinate. For (3, 5), the x-coordinate is 3.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Plot the point (4, 2). The student plotted the point at (2, 4). Is this correct? Student says: Yes

Check the order of the coordinates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The x-coordinate comes first. For (4, 2), go 4 units right and 2 units up, not the other way.	**No. The point is at (4, 2)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the ordered pair for the origin? Student's answer: (1, 1)

Find the origin.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The origin is where the axes meet, at zero on both axes. Its ordered pair is (0, 0).	**(0, 0)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A point is at (6, 0). Where is it located? Student's answer: on the y-axis

Look at the second coordinate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The second number is 0, so the point is on the x-axis, 6 units to the right of the origin.	**On the x-axis**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mia walks 6 units east, then 2 units north. What ordered pair is her new location? Student's answer: (2, 6)

Write the x-coordinate first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
East movement is on the x-axis, so it comes first. Her location is (6, 2).	**(6, 2)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In which quadrant is the point (2, 9)? Student's answer: second quadrant

Look at the signs of the coordinates.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Both numbers are positive, so the point is in the first quadrant.	**First quadrant**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

hours studied on the x-axis and score on the y-axis. Which point shows 2 hours and a score of 60? Student'

Match each value to its axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Hours are on the x-axis, so they come first. The point is (2, 60).	**(2, 60)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, 0) and (4, 8). Is the point (2, 4) on the line? Student says: No

Check whether the point fits the pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The line goes up 2 for every 1 across. (2, 4) fits: 2 is half of 4, and 4 is half of 8. It is on the line.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

t 15 cents. Write the point that represents this, with pencils on the x-axis and cents on the y-axis. Student's a

Put pencils first, then cents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Pencils are on the x-axis, so the point is (3, 15).	**(3, 15)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

on day 1, 5 inches on day 2, and 7 inches on day 3. What point shows day 4 if it grows at the same rate? St

Find the pattern and the next value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The plant grows 2 inches each day. On day 4 it is 7 + 2 = 9 inches, so the point is (4, 9).	**(4, 9)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Where is the point (0, 6) located? Student says: on the x-axis at 6

Look at the coordinates.

What did the student do wrong? Why do you think the student made this error?

The first number is 0, so the point is on the y-axis, 6 units up from the origin.

Solve the problem correctly. Show all of your work.

On the y-axis

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ray says that (5, 0) is on the y-axis. Is he correct? Student says: Yes

Check the coordinates.

What did the student do wrong? Why do you think the student made this error?

The second number is 0, so the point is on the x-axis, not the y-axis.

Solve the problem correctly. Show all of your work.

No. It is on the x-axis

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every square a rectangle? Student says: No

Think about the properties of each shape.

What did the student do wrong? Why do you think the student made this error?

A square has four right angles and two pairs of parallel sides, so it meets the rules of a rectangle. Every square is a rectangle.

Solve the problem correctly. Show all of your work.

Yes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every rectangle a square? Student says: Yes

Think about the properties of each shape.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A square needs all four sides equal. A 4 by 2 rectangle has four right angles but is not a square.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every rhombus a square? Student says: Yes

Think about the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A rhombus has four equal sides, but it does not need right angles. A square needs right angles too.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every trapezoid a parallelogram? Student says: Yes

Think about parallel sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A trapezoid has only one pair of parallel sides. A parallelogram needs two pairs.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every rhombus a parallelogram? Student says: No

Think about parallel sides.

What did the student do wrong? Why do you think the student made this error?

A rhombus has two pairs of parallel sides, which is the rule for a parallelogram. Every rhombus is a parallelogram.

Solve the problem correctly. Show all of your work.

Yes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is every rectangle a parallelogram? Student says: No

Think about parallel sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A rectangle has two pairs of parallel opposite sides, so it is a parallelogram.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

iangle has three sides that are each 3 cm long. How can you classify it by its sides? Student's answer: scal

Classify by the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A scalene triangle has no equal sides. All three sides are equal, so it is equilateral.	**Equilateral**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles of 30°, 60°, and 90°. How can you classify it by its angles? Student's answer: acute

Classify by the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
One angle is exactly 90°, so the triangle is a right triangle. An acute triangle has all angles less than 90°.	**Right triangle**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

has four right angles and two pairs of parallel sides, but the sides are not all equal. What is it? Student's an

Look at the angles and the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A rhombus has four equal sides, but this shape does not. Four right angles and parallel sides make it a rectangle.	**Rectangle**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A quadrilateral has four equal sides, but no right angles. What is it? Student's answer: square

Look at the angles and the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A square needs right angles. A quadrilateral with four equal sides and no right angles is a rhombus.	**Rhombus**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A quadrilateral has exactly one pair of parallel sides. What is it? Student's answer: parallelogram

Count the pairs of parallel sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A parallelogram has two pairs of parallel sides. A quadrilateral with one pair is a trapezoid.	**Trapezoid**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has two sides that are the same length. What is it? Student's answer: scalene

Classify by the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A scalene triangle has no equal sides. Two equal sides make it an isosceles triangle.	**Isosceles triangle**